SOUVIK TA

Toronto, ON souvikat@gmail.com LinkedIn GitHub Medium Portfolio

EXPERIENCE

Machine Learning Engineer Intern
Multiverse Computing • *Toronto, ON* • *Sept 2026 – Present*

- Developing a **time-series forecasting pipeline** for electrical grid data whose predictions feed quantum-inspired network optimisation.
- Designed a **predictive maintenance framework** for an industrial facilities client that flags faults in air handling units, estimates capital costs, and guides repair-versus-replace decisions.

PhD Researcher — AI & Machine Learning
Western University • *London, ON* • *Sept 2021 – Aug 2026*

- Trained PyTorch models that predict industrial reactor behaviour — **40–55% lower prediction error** than standard ML approaches. *(Published, Engineering Applications of Artificial Intelligence)* **[Paper]**
- Engineered data cleaning and validation pipelines for **6 years of real plant data**, feeding a predictive model with uncertainty estimates. *(Published, I&ECR)*
- Proposed a neural network architecture for an industrial refinery unit — **23.5% lower prediction error** than standard ML baselines. *(Published, Fuel)*
- Applied **1D-CNN** models to predict 10 crude oil properties from spectral data. *(Published, Canadian Journal of Chemical Engineering)* **[Paper]**

LLM & GENERATIVE AI PROJECTS

ChemSimGuide — Agentic Process Simulator
Google GenAI Capstone 2025 — Global Top 20 Finalist (6,000+ teams)

- Built an LLM pipeline that runs **RAG** over technical literature, generates code, executes it in **Docker**, and returns visualised results.
- Designed prompts with **structured outputs** (JSON/Pydantic) and **function calling**, plus multi-turn loops that feed errors back for refinement.

ODE Forge — Autonomous Simulation Agent
LLM Hackathon for Material Science and Chemistry

- Created a **LangGraph** agent that turns plain-language problems into self-correcting workflows.

Agentic Wastewater Epidemiology System
MITACS-Funded — II Prize, Best Oral Presentation, IIChE CHEMCON 2025

- Built an LLM workflow that fits models to surveillance data using **prompt templates** and feedback loops.

EDUCATION

PhD, Chemical and Biochemical Engineering Western University, London, ON 2026
Thesis: Machine Learning Frameworks for Hydroprocessing Systems: Data-Driven and Physics-Informed Approaches
B.E., Chemical Engineering BITS Pilani, India 2020

TECHNICAL SKILLS

LLM Engineering	Prompt engineering, structured outputs (JSON/Pydantic), function calling, RAG, LangGraph, LangChain, OpenAI API, vector databases
Data & Deployment	Python, data pipelines, FastAPI, Docker, CI/CD, SQL/PostgreSQL, Git, Azure
Machine Learning	PyTorch, Jupyter, time-series forecasting, model evaluation, uncertainty quantification

LEADERSHIP & WRITING

- AI Technical Writer, Medium (2024–Present) — 14+ articles on GenAI, RAG, and agentic AI
- Director, Board of Directors, Society of Graduate Students, Western University (2024–2025)